

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 12



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Recap of Previous Lecture



- ✓ **Topic** Functions
- ✓ **Topic** Total number of functions
- ✓ **Topic** One-one and onto functions
- ✓ **Topic** Number of one-one functions
- ✓ **Topic** Number of onto functions

Topics to be Covered



✓ Topic

Bijjective function and inverse of a function

✓ Topic

Identity function and constant function

✓ Topic

Identical function

✓ Topic

Function composition

		No. of onto functions from set A to set B
①	$ A =n$ & $ B =2$ ($n \geq 2$)	$2^n - 2$
②	$ A =n$ & $ B =n$	$n!$
③	$ A =n$ & $ B =n-1$ $ A = 1 + B $	${}^nC_2 \cdot (n-1)!$
④	$ A =m$ & $ B =n$ Where ($m \geq n$)	$\sum_{i=0}^{(n-1)} (-1)^i \cdot {}^nC_i \cdot (n-i)^m$

Today's Topic



Topic : Bijective Function

A function $f: A \rightarrow B$ is called a bijective function if and only if,

- (i) $f: A \rightarrow B$ is one-one
- & (ii) $f: A \rightarrow B$ is onto.

$f: A \rightarrow B$ is bijective iff $f: A \rightarrow B$ is one-one & $f: A \rightarrow B$ is onto

i.e., $|A| \leq |B|$ ✓ & $|A| \geq |B|$ ✓

A bijective function is possible from set A to set B only if $|A| = |B|$

if, $|A| = |B| = n$,

then Number of bijective function
Possible from set A to set B = $n!$

Note:- ① If bijective function exist between the sets A and B,
then Cardinality of set A = Cardinality of set B.

② If bijective function exist between the sets A and B,
then set A and set B are said to have
One-one Correspondance.
(different from one-one function)

Q: Let $R =$ Set of all real numbers
& $N =$ Set of all natural numbers

We can not define any bijective function b/w
Set of all real nos & Set of all natural nos.

∴ We can say that R & N are not Equivalent

Q: $A = \{0, 2, 4, 6, 8, 10, \dots\}$

$$N = \{0, 1, 2, 3, 4, 5, \dots\}$$

We can define a function $f: N \rightarrow A$

s.t. $f(x) = 2x$

Here 'f' is
One-one as well
as onto.

i.e., f is bijective

We can define a bijective function b/w A & N.

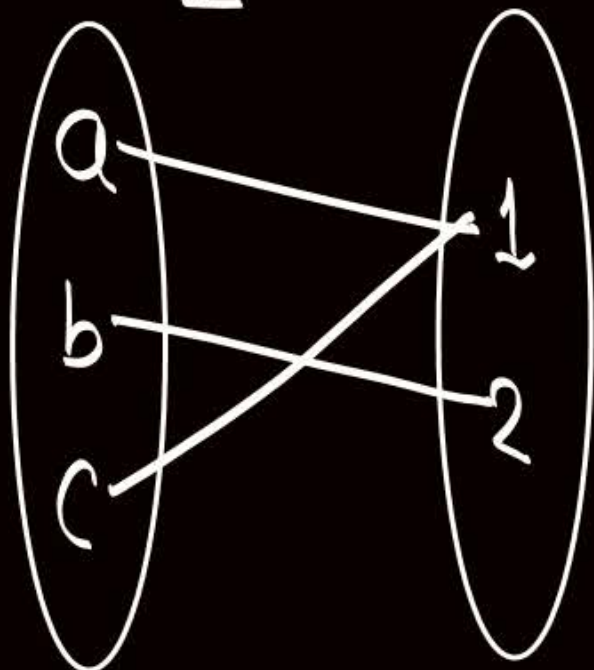
∴ Set A & Set N are equivalent



Topic : Inverse of a Function

- Inverse of a function "f" can exist if and only if, function "f" is a bijective function
- If $f: A \rightarrow B$ is a bijective function, then inverse function w.r.t. function 'f' can be denoted by f^{-1} ,
And it will be defined from set B to set A.

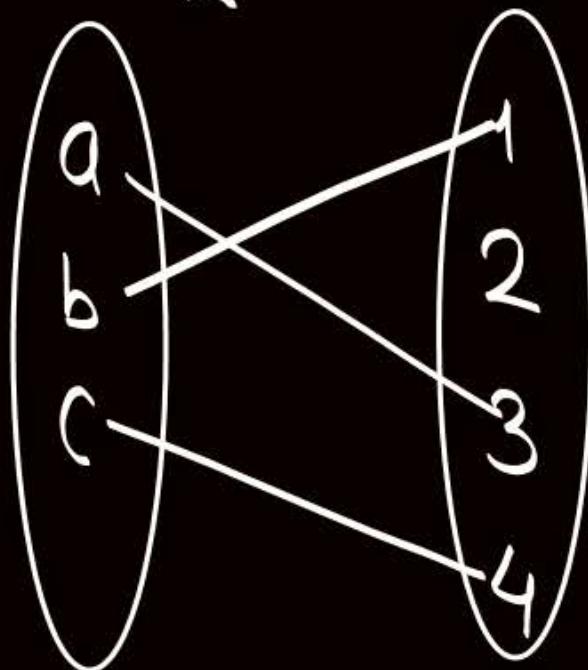
$A \quad f_1: A \rightarrow B \quad B$



f_1 is onto,
but not one-one

→ Inverse relation wrt.
function f_1 will not
be a function

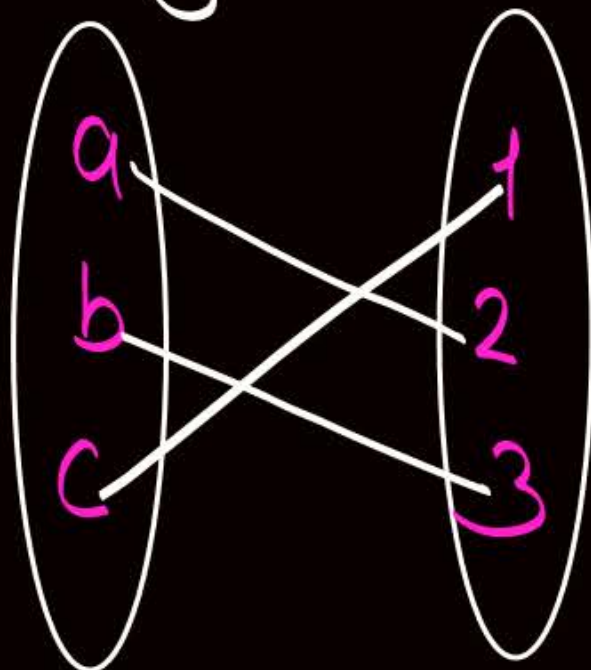
$A \quad f_2: A \rightarrow B \quad B$



f_2 is one-one
but not onto

→ Inverse relation wrt.
function f_2 will not
be a function

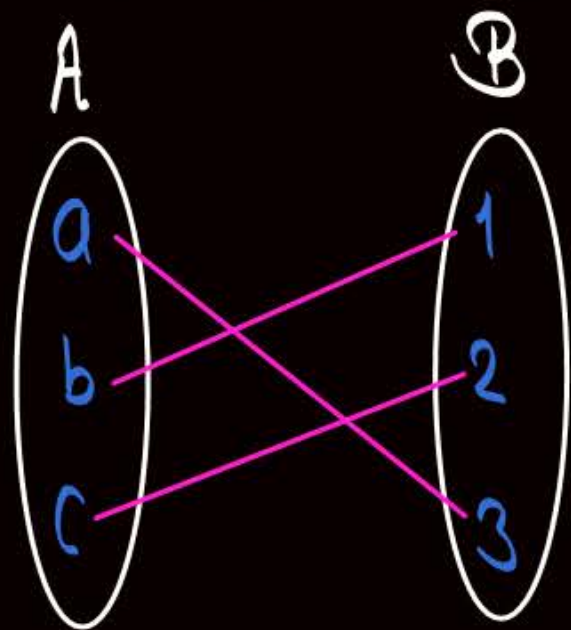
$A \quad f_3: A \rightarrow B \quad B$



f_3 is one-one
as well as onto

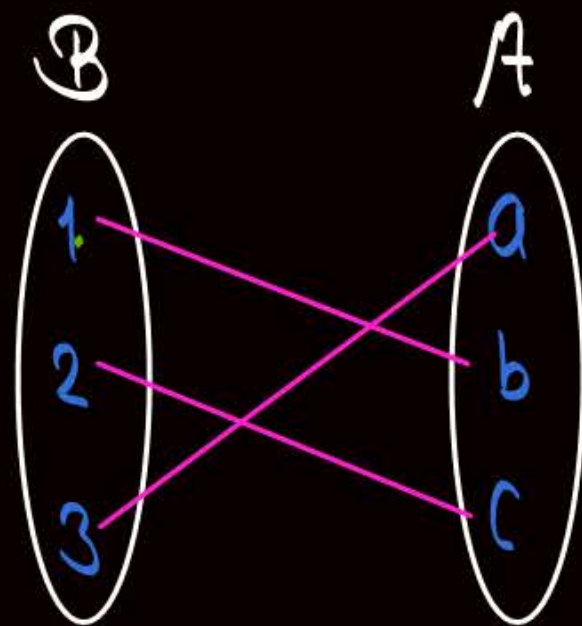
→ Inverse relation wrt.
function f_3
will be a function

eg. let $f: A \rightarrow B$ is bijection $\therefore$ (i.e. invertible)



$$\begin{aligned} f(a) &= 3 \\ f(b) &= 1 \\ f(c) &= 2 \end{aligned}$$

$f^{-1}: B \rightarrow A$



$$\begin{aligned} f^{-1}(3) &= a \\ f^{-1}(1) &= b \\ f^{-1}(2) &= c \end{aligned}$$

i.e. if $f(x) = y$
then $f^{-1}(y) = x$

Q: Let 'f' is a bijective function

defined as $f(x) = 2x + 3$

Obtain the inverse function w.r.t. function 'f'.

i.e., $f^{-1}(x) = ?$

$$\text{Let } f(x) = y \implies 2x + 3 = y$$

$$\therefore \underbrace{f^{-1}(y)} = x$$

f^{-1} is a function of y
 $\therefore$ We must represent ' x '
in terms of y

$$\therefore x = \left(\frac{y-3}{2} \right)$$

$$\text{Hence } f^{-1}(y) = \left(\frac{y-3}{2} \right) \implies$$

$$f^{-1}(x) = \frac{x-3}{2}$$

Gate 1996 :

Q. Let, $f((x,y)) = (x+y, x-y)$ }
find $f^{-1}((x,y)) = ?$

function 'f' is defined

$$f: \underbrace{\mathbb{R} \times \mathbb{R}}_{\text{Domain}} \rightarrow \underbrace{\mathbb{R} \times \mathbb{R}}_{\text{Codomain}}$$

Domain

it is a set
of all order
pairs of
real numbers

Codomain

it is a set
of all order
pairs of
real numbers

Gate 1996 :

Q. Let, $f(x, y) = (x+y, x-y)$

find $f^{-1}(x, y)$

$$\text{Let } f(x, y) = (a, b) \implies (x+y, x-y) = (a, b)$$

$$\therefore f^{-1}(a, b) = (x, y)$$

$$= \left(\frac{a+b}{2}, \frac{a-b}{2} \right)$$

$$f^{-1}(x, y) = \left(\frac{x+y}{2}, \frac{x-y}{2} \right)$$

$$\therefore x+y = a \text{ --- ①}$$

$$\& x-y = b \text{ --- ②}$$

By eqⁿ ① & eqⁿ ②

$$\left(x = \frac{a+b}{2} \right) \& \left(y = \frac{a-b}{2} \right)$$



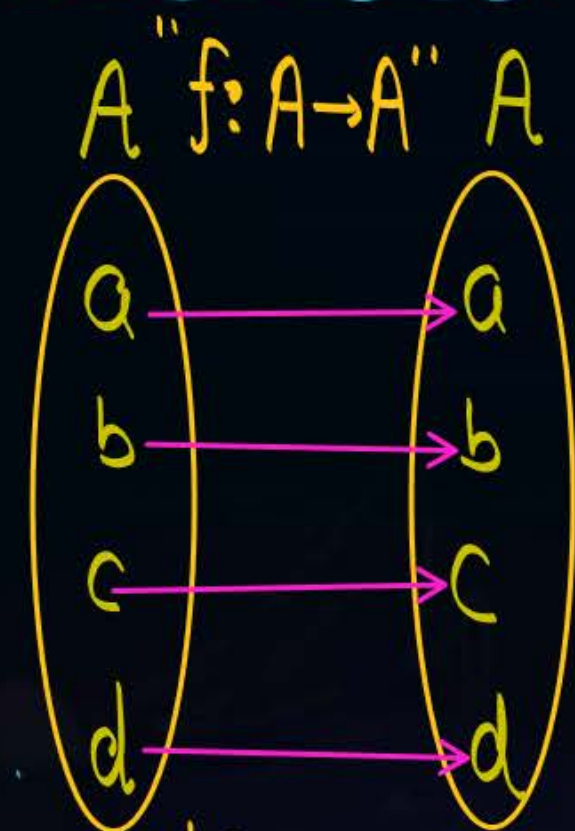
Topic : Identity Function

A function "f" from set A to set A (i.e. on set A) is called an identity function if and only if $f(x) = x, \forall x \in A$

• Generally, Identity function on set A is denoted by I_A .

* Every identity function is always a bijection

* Inverse of identity function will always be that identity function itself i.e. $I_A^{-1} = I_A$



Here 'f' is an identity function



Topic : Constant Function

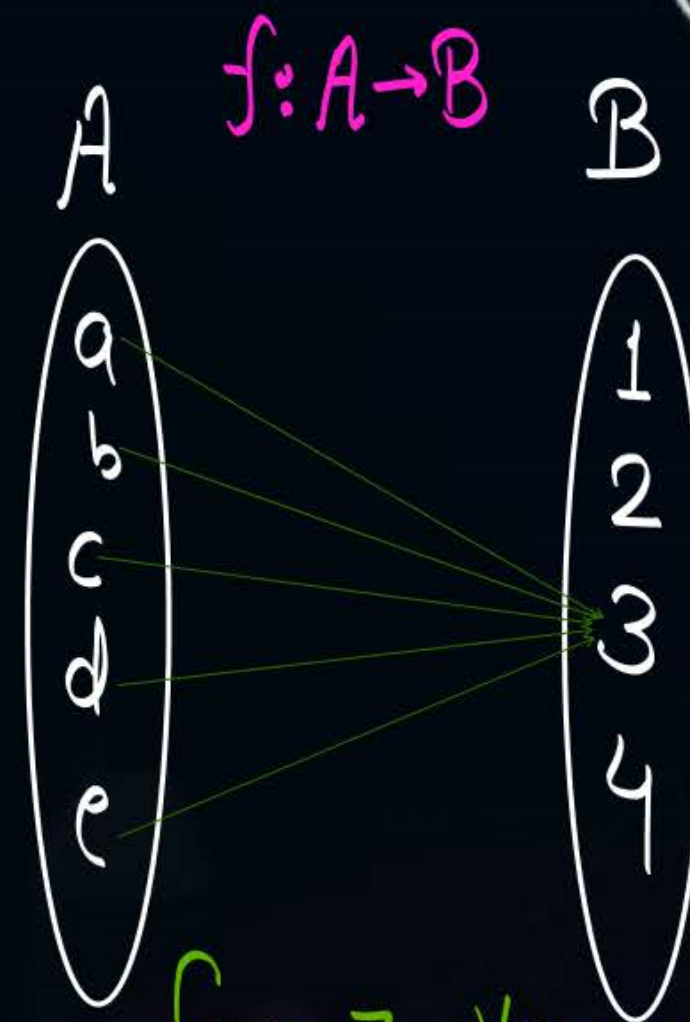
A function $f: A \rightarrow B$
is called a Constant function,
if and only if, $\underline{f(x) = c}, \forall x \in A$

A fixed value
from Co-domain

• If $|A|=m$ and $|B|=n$, then

No. of Constant functions possible

from set A to set B $= n =$ Cardinality of Co-domain



$f(x) = 3, \forall x \in A$
fixed value
from Co-domain
∴ It is a
Constant function



Topic : Identical function

Equal functions



* Two functions 'f' and 'g' are said to be identical if and only if,

- ✓ ① Domain of f = Domain of g
- and ✓ ② Range of f = Range of g
- and ✓ ③ $f(x) = g(x) \quad \forall x \in \text{Domain}$

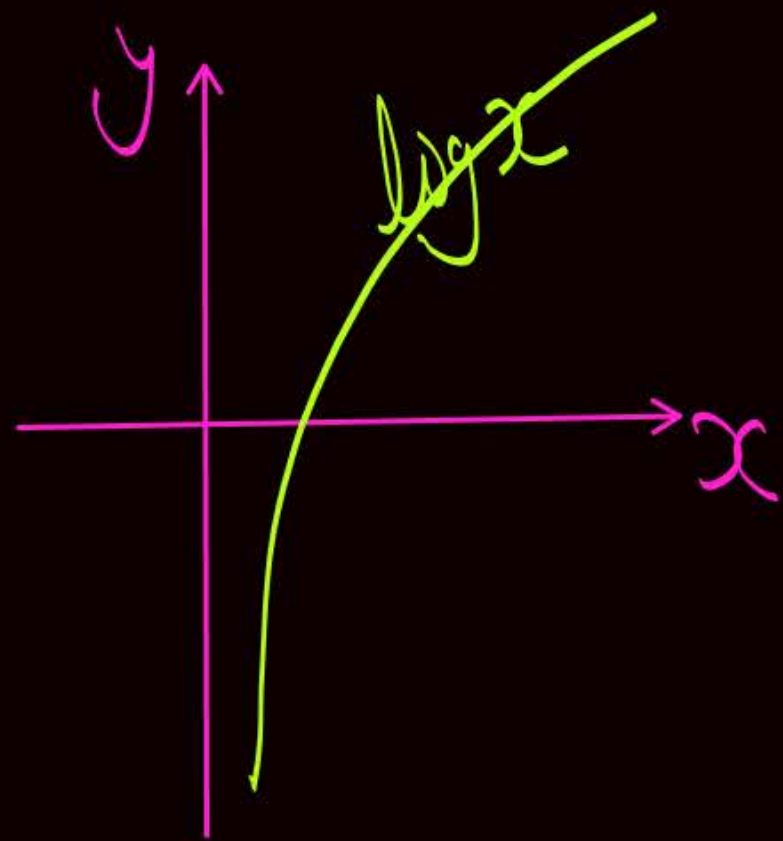
Q:- Check whether the functions 'f' and 'g' are identical or not?

$$f = \log(x^2)$$

Here $x^2 > 0 \therefore x \in \mathbb{R} - \{0\}$

$$g = 2 \log(x)$$

Here $x > 0 \therefore x \in \mathbb{R}^+$



Domain of $f \neq$ Domain of g
 $\therefore f$ & g are not identical



Topic : Function composition

- Let 'f' and 'g' are two functions, Composite function of 'f' and 'g' (if exist) can be denoted by "fog"
- In general, $f \circ g \neq g \circ f$ } function Composition is not commutative
- $(f \circ g)(x) = f(g(x))$
- $(g \circ f)(x) = g(f(x))$

$$\bullet ((f \circ g) \circ h)$$

$$((f \circ g) \circ h)(x)$$

$$(f \circ g)(h(x))$$

$$f(g(h(x)))$$

$$(f \circ (g \circ h))$$

$$(f \circ (g \circ h))(x)$$

$$f((g \circ h)(x))$$

$$f(g(h(x))) \overset{\text{Equal}}{=} f(g(h(x)))$$

↳ $\circ$ function composition is
Associative (Provided it is defined)

$f: A \rightarrow A$
 $g: A \rightarrow A$
 $g \circ f: A \rightarrow A$
 $g(f(x)) : \text{Domain of } f \rightarrow \text{Co-domain of } g$

$f \circ g: A \rightarrow A$
 $f(g(x))$
 $\uparrow$ domain of g
 $\uparrow$ Co-domain of f

$f: A \rightarrow B$
 $g: B \rightarrow A$
 $g \circ f: A \rightarrow A$
 $g(f(x)) : \text{Domain of } f \rightarrow \text{Co-domain of } g$

$f \circ g: B \rightarrow B$
 $f(g(x))$
 $\uparrow$ domain of g
 $\uparrow$ Co-domain of f

$f: A \rightarrow B$
 $g: B \rightarrow C$

$g \circ f: A \rightarrow C$
 $g(f(x)) : \text{Domain of } f \rightarrow \text{Co-domain of } g$

$f \circ g$: Need not be defined
 $f(g(x))$
 $\Downarrow$ is defined
 Only if
 $\text{Range of } g \subseteq \text{Domain of } f$

* Let $f: A \rightarrow A$ is a bijective function



$\hookrightarrow \therefore f^{-1}$ is also defined
 $f^{-1}: A \rightarrow A$

$$f \circ f^{-1}: A \rightarrow A$$

$$(f \circ f^{-1})(x) = x$$

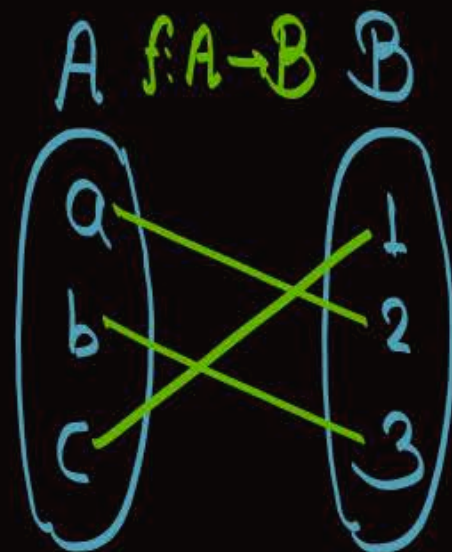
$$\therefore f \circ f^{-1} = I_A$$

$$f^{-1} \circ f: A \rightarrow A$$

$$(f^{-1} \circ f)(x) = x$$

$$f^{-1} \circ f = I_A$$

* Let $f: A \rightarrow B$ is a bijective function



$\hookrightarrow \therefore f^{-1}$ is also defined
 $f^{-1}: B \rightarrow A$

$$f \circ f^{-1}: B \rightarrow B$$

$$(f \circ f^{-1})(x) = x$$

$$f \circ f^{-1} = I_B$$

$$f^{-1} \circ f: A \rightarrow A$$

$$(f^{-1} \circ f)(x) = x$$

$$f^{-1} \circ f = I_A$$

→ let $f: A \rightarrow A$

$$f \circ I_A: A \rightarrow A$$

$$(f \circ I_A)(x) = f(I_A(x)) \\ = f(x)$$

$$f \circ I_A = f$$

$$I_A \circ f: A \rightarrow A$$

$$(I_A \circ f)(x) = I_A(\underbrace{f(x)}_{\text{value from set A}}) \\ = f(x)$$

$$I_A \circ f = f$$

→ let $f: A \rightarrow B$

$$f \circ I_A: A \rightarrow B$$

$$f \circ I_A = f$$

$I_A \circ f$: Need not be defined

$f \circ I_B$: Need not be defined

$$I_B \circ f: A \rightarrow B$$

$$I_B \circ f = f$$

Note:-

If $*$ is a binary operation, then following property will always hold true

$$(a * b)^{-1} = b^{-1} * a^{-1}$$

does not matter, whether opⁿ is commutative or not

But, we can say

$$(a * b)^{-1} = a^{-1} * b^{-1} \text{ will also hold true}$$

if we already know that $*$ is commutative

And it need not hold true if ' $*$ ' is not commutative

→ function composition is not commutative

∴ $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$ will always hold true

but $(f \circ g)^{-1} = f^{-1} \circ g^{-1}$ need not be true

→ W.r.t. Matrix multiplication

$(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$ will always hold true

But $(A \cdot B)^{-1} = A^{-1} \cdot B^{-1}$ need not hold true

Q.42	Let $\mathcal{F}$ be the set of all functions from $\{1, \dots, n\}$ to $\{0, 1\}$. Define the binary relation $\preceq$ on $\mathcal{F}$ as follows: $\forall f, g \in \mathcal{F}, f \preceq g$ if and only if $\forall x \in \{1, \dots, n\}, f(x) \leq g(x)$, where $0 \leq 1$. Which of the following statement(s) is/are TRUE?
(A)	$\preceq$ is a symmetric relation
(B)	$(\mathcal{F}, \preceq)$ is a partial order
(C)	$(\mathcal{F}, \preceq)$ is a lattice
(D)	$\preceq$ is an equivalence relation

HW Check whether the statement is True or False?

① If $fo g$ is onto then ' f ' must be onto

② If ' f ' is onto then ' $fo g$ ' is onto

③ If ' f ' is one-one then $fo g$ is one-one

④ If ' $fo g$ ' is one-one then ' f ' must be one-one

⑤ If ' $fo g$ ' is one-one, then ' g ' must be one-one

Q.17

$g(.)$ is a function from A to B , $f(.)$ is a function from B to C , and their composition defined as $f(g(.))$ is a mapping from A to C .

If $f(.)$ and $f(g(.))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(.)$?

(A)

$g(.)$ must be an onto (surjective) function.

(B)

$g(.)$ must be a one-to-one (injective) function.

(C)

$g(.)$ must be a bijective function, that is, both one-to-one and onto.

(D)

$g(.)$ is not required to be a one-to-one or onto function.



2 mins Summary



Topic

Bijjective function and inverse of a function

Topic

Identity function and constant function

Topic

Identical function

Topic

Function composition

THANK - YOU